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Inventor(s)	Kobayashi; Masahiro et al.

Torque measuring device

Abstract

A torque measuring device includes a bridge circuit in which four detection coils arranged around a magnetostrictive effect section of a rotating shaft are arranged on four sides; and the bridge circuit includes a resistance element connected to at least one of the four sides for adjusting a resistance value of the side.

Inventors: **Kobayashi; Masahiro (Fukuroi, JP), Fukuda; Kota (Fukuroi, JP), Odera; Takahiro (Fukuroi, JP), Futakuchi; Naoki (Koto-ku, JP)**

Applicant: **NSK LTD.** (Tokyo, JP); **Proterial, Ltd.** (Tokyo, JP)

Family ID: **87851343**

Assignee: **NSK LTD. (Tokyo, JP); PROTERIAL, LTD. (Tokyo, JP)**

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Primary Examiner: Kirkland, III; Freddie

Attorney, Agent or Firm: Sughrue Mion, PLLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
(1) This patent application claims the benefit of JP Patent Application No. 2022-031026 filed Mar. 1, 2022. The above application is incorporated by reference herein.
FIELD
(2) The present disclosure relates to a torque measuring device capable of measuring torque transmitted by a rotating shaft.
BACKGROUND
(3) In recent years, in the field of automobiles, the development of systems that measure torque transmitted by a rotating shaft of a power train, or in other words, a power transmission mechanism, use the measurement results to control output of an engine or an electric motor that is a power source, and execute speed change control of a transmission is advancing.
(4) Conventionally, a magnetostrictive torque measuring device is known as a device for measuring torque transmitted by a rotating shaft. A magnetostrictive torque measuring device is configured to measure torque being transmitted by a rotating shaft having a magnetostrictive effect section whose

magnetic permeability changes when torque is applied, and detecting the change in magnetic permeability of the magnetostrictive effect section when torque is applied as a change in the inductance of a detection coil.

(5) In general, the change in magnetic permeability of the magnetostrictive effect section due to fluctuation in torque is very small, and therefore, measures are usually taken to improve the measurement sensitivity. As one such measure, a method of using a bridge circuit **100** as illustrated in FIG. **9** is known as disclosed in JP 2016-200552 A, JP 2017-049124 A, or the like. In this method, as illustrated in FIG. **10**, for example, a first detection coil **103**, a second detection coil **104**, a third detection coil **105**, and a fourth detection coil **106**, which are four detection coils of a bridge circuit **100**, are arranged around a columnar magnetostrictive effect section **102**, which is a part of a rotating shaft **101** in the axial direction.

(6) When a torque T is applied to the rotating shaft **101**, stresses **6** with different positive and negative signs act on an outer peripheral surface of the magnetostrictive effect section **102** in directions inclined +45° and −45° with respect to the axial direction. Due to an inverse magnetostriction effect, the magnetic permeability increases in a direction in which the tensile stress (+σ) acts, and decreases in the direction in which the compressive stress (−σ) acts.

(7) In the bridge circuit **100**, the first detection coil **103** (inductance L**1**) and the third detection coil **105** (inductance L**3**) arranged as one set of two sets of opposite sides forming the four sides are arranged on the outer peripheral surface of the magnetostrictive effect section **102**, and are detection coils for detecting a change in magnetic permeability in a direction inclined +45° with respect to the axial direction, and the second detection coil **104** (inductance L**2**) and the fourth detection coil **106** (inductance L**4**) arranged as another set of two sets of opposite sides are arranged on the outer peripheral surface of the magnetostrictive effect section **102**, and are detection coils for detecting a change in magnetic permeability in a direction inclined −45° with respect to the axial direction.

(8) In such a bridge circuit **100**, when an input voltage Vi is applied between two end points, points A and C, an output voltage Vo corresponding to a direction and a magnitude of torque T applied to the rotating shaft **101** is obtained as voltage between two midpoints, points B and D. Therefore, the torque T can be measured based on the output voltage Vo.

(9) By using the bridge circuit **100** as described above, it is possible to measure the torque T with twice the sensitivity as compared with a case in which torque T is measured by detecting only the change in magnetic permeability in one of the +45° inclined direction and −45° inclined direction with respect to the axial direction.

SUMMARY

(10) However, in general, each of the first detection coil **103**, the second detection coil **104**, the third detection coil **105**, and the fourth detection coil **106** has resistance components r**1**, r**2**, r**3**, and r**4** represented by copper wire resistance, and variations in these resistance components r**1**, r**2**, r**3**, and r**4** affect the output voltage Vo of the bridge circuit **100**. More specifically, these resistance components r**1**, r**2**, r**3**, and r**4** increase and decrease with changes in temperature, that is, with changes in the electrical resistivity of copper, and the output voltage Vo also increases and decreases with the increase and decrease of the resistance components r**1**, r**2**, r**3**, and r**4**.

(11) Then, the amount of change in the output voltage Vo with respect to the temperature change at this time increases as the variations in the resistance components r**1**, r**2**, r**3**, and r**4**, that is, the variations in the resistance values of the four sides of the bridge circuit **100** increase.

(12) An object of the present disclosure is to provide a torque measuring device capable of suppressing changes in output voltage of a bridge circuit due to temperature changes.

(13) A torque measuring device according to an aspect of the present disclosure includes a bridge circuit in which four detection coils arranged around a magnetostrictive effect section of a rotating shaft are arranged on four sides. The bridge circuit includes a resistance element connected to at least one of the four sides for adjusting a resistance value of the side.

(14) In the torque measuring device according to an aspect of the present disclosure, a resistance element is connected to each of the four sides.
(15) In the torque measuring device according to an aspect of the present disclosure, the resistance value of the resistance element connected to the at least one of the four sides is adjusted so that the ratio (R**1**×R**3**)/(R**2**×R**4**) of a product R**1**×R**3** of resistance values R**1** and R**3** of two opposite sides forming one pair of opposite sides of the four sides, and a product R**2**×R**4** of resistance values R**2** and R**4** of two opposite sides forming another pair of opposite sides of the four sides approaches 1 compared to a case in which the resistance element is not connected to each of the four sides.
(16) In the torque measuring device according to an aspect of the present disclosure, the resistance element has a configuration in which the resistance value between end points on both sides of the resistance element, which is the resistance value of the resistance element, is adjustable by changing a path length of copper wiring connecting the end points on both sides of the resistance element.
(17) In the torque measuring device according to an aspect of the present disclosure, the resistance element has a configuration in which a plurality of parallel copper wires are connected in parallel in the middle of the copper wiring connecting the end points on both sides of the resistance element, and the resistance value between the end points on both sides, which is the resistance value of the resistance element, is adjustable by changing the number of the plurality of parallel copper wires by cutting a part of the plurality of parallel copper wires.
(18) The torque measuring device according to an aspect of the present disclosure includes a coil unit formed in a cylindrical shape and having the four detection coils, a back yoke formed in a cylindrical shape and arranged coaxially around the coil unit, and a holder configured to hold the coil unit and the back yoke.
(19) With the torque measuring device according to an aspect of the present disclosure, it is possible to suppress changes in output voltage of a bridge circuit due to temperature changes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) To assist those of ordinary skill in the relevant art in making and using the Subject matter hereof, reference is made to the appended drawings, in which like reference numerals refer to similar elements.
(2) FIG. **1** is a cross-sectional view of a torque measuring device according to a first example of an embodiment of the present disclosure.
(3) FIG. **2** is a diagram illustrating a bridge circuit of the torque measuring device of the first example.
(4) FIG. **3** is a developed view of a flexible substrate of a coil unit of the torque measuring device of the first example.
(5) FIG. **4** is a developed view of a first detection coil, a second detection coil, a third detection coil, and a fourth detection coil of the coil unit of the torque measuring device of the first example, as viewed from the outside in the radial direction.
(6) FIG. **5**A to FIG. **5**D respectively illustrate a first detection coil, a second detection coil, a third detection coil, and a fourth detection coil of the coil unit of the torque measuring device of the first example, and are developed views of each single unit thereof viewed from the outside in the radial direction.
(7) FIG. **6** is a diagram illustrating a first configuration example of resistance elements connected respectively to four sides of the bridge circuit of the torque measuring device of the first example.
(8) FIG. **7** is a diagram illustrating a second configuration example of resistance elements connected respectively to each of the four sides of the bridge circuit of the torque measuring device

of the first example.
(9) FIG. **8** is a diagram corresponding to FIG. **2** in a second example of an embodiment of the present disclosure.
(10) FIG. **9** is a diagram illustrating a bridge circuit of a conventional torque measuring device.
(11) FIG. **10** is a perspective view of a rotating shaft for explaining a direction of stress generated when torque is applied to the rotating shaft.
(12) It should be understood that the drawings are not to scale and that the disclosed embodiments are sometimes illustrated diagrammatically and in partial views. In certain instances, details that are not necessary for an understanding of the disclosed method and apparatus, or that would render other details difficult to perceive may have been omitted. It should be understood that this disclosure is not limited to the particular embodiments illustrated herein.

DETAILED DESCRIPTION

(13) In the following detailed description of some embodiments, reference is made to the accompanying drawings, which form a part hereof, and within which are shown by way of illustration specific embodiments by which the disclosure may be practiced. It is to be understood that other embodiments may be utilized, and structural changes may be made without departing from the scope of the disclosure.

First Example

(14) A first example of an embodiment of the present disclosure will be described with reference to FIGS. **1** to **7**.
(15) A torque measuring device **1** of this example is a device for measuring the torque transmitted by a rotating shaft **2**, and can be used by being incorporated in various mechanical devices. Specific examples of a mechanical device incorporating the torque measuring device **1** of the present example include: a mechanical device of a power train of an automobile, for example, a transmission such as an automatic transmission (AT), a belt-type continuously variable transmission, a toroidal type continuously variable transmission, an automatic manual transmission (AMT), a dual clutch transmission (DCT) or the like that performs gear shifting that is controlled on the vehicle side; or a transfer, a manual transmission (MT), or the like. The driving system of the target vehicle is not particularly limited and may be FF, FR, MR, RR, 4WD, or the like.
(16) Specific examples of the other mechanical devices incorporating the torque measuring device **1** of the present example include devices that change a rotation speed of a power shaft with gears such as a speed reducer or a speed increaser of a wind turbine, a railway vehicle, a rolling mill for steel, and the like.
(17) In this example, the rotating shaft **2** is a rotating shaft incorporated in a mechanical device of a power train as described above, and is rotatably supported by a rolling bearing (not illustrated) with respect to a casing (not illustrated) that does not rotate during use, and has a magnetostrictive effect section, the magnetic permeability of which changes according to torque to be transmitted.
(18) The rotating shaft **2** has an intermediate shaft portion **3** at an intermediate portion in an axial direction thereof as illustrated in FIG. **1**. An outer peripheral surface of the intermediate shaft portion **3** is configured by a cylindrical surface. In this example, the intermediate shaft portion **3** of the rotating shaft **2** functions as a magnetostrictive effect section. For this reason, the rotating shaft **2** is made of a magnetic metal. As the magnetic metal forming the rotating shaft **2**, various magnetic steels can be used such as carburized steel such as SCr420 and SCM420, and carbon steel such as S45C, which are defined in the Japanese Industrial Standards (JIS).
(19) When a torque T is applied to the rotating shaft **2**, stresses **6** with differing positive and negative signs act on the outer peripheral surface of the intermediate shaft portion **3** in directions inclined +45° and −45° with respect to the axial direction. Due to an inverse magnetostriction effect, the magnetic permeability increases in a direction in which the tensile stress (+σ) acts, and decreases in the direction in which the compressive stress (−σ) acts.
(20) When carrying out the present disclosure, it is possible to improve the mechanical and

magnetic properties of a portion of the outer peripheral surface of the intermediate shaft portion **3** around which the detection coils **5** to **8** of the torque measuring device **1** are arranged by subjecting the portion to a shot peening process to form a compression work hardened layer. In this way, sensitivity and hysteresis of torque measurement by the torque measuring device **1** may be improved.

(21) When carrying out the present disclosure, instead of having the intermediate shaft portion **3** function as the magnetostrictive effect section, it is also possible to fix a magnetostrictive material functioning as the magnetostrictive effect section to an outer peripheral surface of the intermediate shaft portion **3**. More specifically, an annular-shaped magnetostrictive material may be fitted around the intermediate shaft portion **3** and fixed, or a magnetostrictive material composed of a film coating such as plating or a film-like magnetostrictive material may be fixed to the outer peripheral surface of the intermediate shaft portion **3**.

(22) The torque measuring device **1** of this example includes a bridge circuit **4** as illustrated in FIG. **2**. On the four sides of the bridge circuit **4**, four detection coils composed of a first detection coil **5**, a second detection coil **6**, a third detection coil **7**, and a fourth detection coil **8**, are arranged around the intermediate shaft portion **3**.

(23) The bridge circuit **4** has two end points, point A and point C, and two midpoints, point B and point D. The first detection coil **5** is arranged on the side between point D and point A. The second detection coil **6** is arranged on the side between point A and point B. The third detection coil **7** is arranged on the side between point B and point C. The fourth detection coil **8** is arranged on the side between point C and point D.

(24) In the bridge circuit **4**, one of two sets of opposite sides of the four sides, that is, the first detection coil **5** and the third detection coil **7** arranged on the side between points D and A and on the side between points B and C are detection coils on the outer peripheral surface of the intermediate shaft portion **3** for detecting a change in magnetic permeability in a direction inclined +45° with respect to the axial direction, or in other words, are detection coils in which respective inductances L**1** and L**3** thereof change as magnetic permeability in that direction changes.

(25) Another of the two sets of opposite sides of the four sides, that is, the second detection coil **6** and the fourth detection coil **8** arranged on the side between points A and B and on the side between points C and D are detection coils on the outer peripheral surface of the intermediate shaft portion **3** for detecting a change in magnetic permeability in a direction inclined −45° with respect to the axial direction, or in other words, are detection coils in which respective inductances L**2** and L**4** thereof change as magnetic permeability in that direction changes.

(26) The bridge circuit **4** has a resistance element connected to at least one side of the four sides for adjusting the resistance value of the side. In this example, a resistance element is connected to each of the four sides of the bridge circuit **4**. That is, in this example, the bridge circuit **4** includes a first resistance element **9**, a second resistance element **10**, a third resistance element **11** and a fourth resistance element **12** as the four resistance elements connected to each of the four sides for adjusting the resistance value of the sides.

(27) More specifically, in this example, the first resistance element **9** is connected in series with the first detection coil **5** on the side between the point D and point A. The second resistance element **10** is connected in series with the second detection coil **6** on the side between the point A and point B. The third resistance element **11** is connected in series with the third detection coil **7** on the side between the point B and point C. The fourth resistance element **12** is connected in series with the fourth detection coil **8** on the side between the point C and point D. However, when carrying out the present disclosure, a resistance element may also be connected in parallel with the detection coil on each side of the bridge circuit.

(28) In the bridge circuit **4**, the resistance value of each of the four sides can be adjusted by changing the resistance value of the resistance element connected to the side. More specifically, the resistance value R**1** between the points D and A can be adjusted by changing the resistance value of

the first resistance element **9**. The resistance value R**2** between the points A and B can be adjusted by changing the resistance value of the second resistance element **10**. The resistance value R**3** between the points B and C can be adjusted by changing the resistance value of the third resistance element **11**. The resistance value R**4** between the points C and D can be adjusted by changing the resistance value of the fourth resistance element **12**.

(29) In this example, in the bridge circuit **4**, the resistance values of resistance elements **9** to **12** connected to each of the four sides are adjusted so that the ratio (R**1**×R**3**)/(R**2**×R**4**) of the product R**1**×R**3** of the resistance values R**1** and R**3** of two opposite sides of the four sides, that form a set of opposite sides, in other words, the side between points D and A and the side between points B and C, and the product R**2**×R**4** of the resistance values R**2** and R**4** of two opposite sides of the four sides, that form another set of opposite sides, in other words, the side between points A and B and the side between points C and D approaches 1 compared to a case where the resistance elements **9** to **12** are not connected to each of the four sides.

(30) For this reason, in this example, before connecting the resistance elements **9** to **12** to the four sides of the bridge circuit **4**, resistance values of respective resistance components r**1**, r**2**, r**3**, and r**4** of the first detection coil **5**, the second detection coil **6**, the third detection coil **7**, and the fourth detection coil **8** are measured with a tester. The resistance values of the resistance components r**1**, r**2**, r**3**, and r**4** that are measured in this way, and the resistance values of the resistance elements **9** to **12** are combined, and the resistance values of the resistance elements **9** to **12** are adjusted so that the ratio (R**1**×R**3**)/(R**2**×R**4**) approaches 1.

(31) For example, the resistance values of the resistance elements **9** to **12** are adjusted so that, of the detection coils **5** to **8** on the four sides, the larger the resistance values of the resistance components r**1**, r**2**, r**3**, and r**4** measured by the tester, the smaller the resistance value of the resistance element connected to the detection coil, and so that the ratio (R**1**×R**3**)/(R**2**×R**4**) approaches 1. Alternatively, the resistance values of the resistance elements **9** to **12** are adjusted so that the resistance values of two or three of the resistance elements **9** to **12** are set to the same magnitude, and the resistance values of the remaining resistance elements are set so that the ratio (R**1**×R**3**)/(R**2**×R**4**) approaches 1.

(32) Note that the resistance values of the resistance elements **9** to **12** may be adjusted by other methods. After adjusting the resistance values as described above, the resistance elements **9** to **12** are connected to the four sides of the bridge circuit **4**.

(33) The bridge circuit **4** includes an oscillator (not illustrated) that applies an input voltage Vi, which is an AC voltage between two end points, point A and point C, and a voltage measuring portion (not illustrated) that detects an output voltage Vo, which is a voltage between two midpoints, point B and point D. In the bridge circuit **4**, when an input voltage Vi is applied between the two end points, point A and point C, by the oscillator, an output voltage Vo corresponding to a direction and a magnitude of the torque T applied to the rotating shaft **2** is obtained as a voltage between the two midpoints, point B and point D. Therefore, when this output voltage Vo is measured by the voltage measuring portion, the direction and magnitude of the torque T can be obtained based on the measured output voltage Vo.

(34) The torque measuring device **1** of this example has an annular shape as a whole, and is supported and fixed to the casing while being arranged coaxially around the intermediate shaft portion **3**. The torque measuring device **1** of this example includes a coil unit **13**, a back yoke **14**, and a holder **15**.

(35) In the present example, the first detection coil **5**, the second detection coil **6**, the third detection coil **7**, the fourth detection coil **8**, and the first resistance element **9**, the second resistance element **10**, the third resistance element **11**, and the fourth resistance element **12** of the bridge circuit **4** are provided in the coil unit **13**. In this example, the coil unit **13** is configured in a cylindrical shape as a whole by a flexible substrate (FPC) having a base film and printed wiring (conductors) held by the base film, and is arranged coaxially around the intermediate shaft portion **3** of the rotating shaft

2.
(36) More specifically, in this example, the coil unit **13** is formed by rolling a band-shaped flexible substrate **16** as illustrated in FIG. **13** into a cylindrical shape, and by joining together, for example, both end portions in the length direction of the flexible substrate **16**. The flexible substrate **16** has four wiring layers that are layered in the thickness direction, and the first detection coil **5**, the second detection coil **6**, the third detection coil **7**, and the fourth detection coil **8**, each of which is configured by printed wiring, are arranged on these wiring layers.
(37) In a state in which the band-shaped flexible substrate **16** is rolled into a cylindrical shape, that is, in a state in which the cylindrical coil unit **13** is formed, the detection coils **5** to **8** are arranged in order of the first detection coil **5**, the second detection coil **6**, the fourth detection coil **8**, and the third detection coil **7** from the inner side in the radial direction.
(38) FIG. **4** illustrates a developed view of the first detection coil **5**, the second detection coil **6**, the third detection coil **7**, and the fourth detection coil **8** as seen from the outer side in the radial direction of the coil unit **13**. FIG. **5**A to FIG. **5**D illustrate developed views of the detection coils **5** to **8** in a single state as seen from the outside in the radial direction of the coil unit **13**.
(39) As illustrated in FIG. **5**A, the first detection coil **5** includes a plurality of coil pieces **17** arranged side by side at equal pitches in a circumferential direction. These coil pieces **17** have a parallelogram shape when viewed from the radial direction, and include wiring that is inclined −45° with respect to the axial direction of the intermediate shaft portion **3**. Coil pieces **17** that are adjacent in the circumferential direction are connected in series by a conductor such as printed wiring (not illustrated).
(40) In FIG. **5**A, the coil pieces **17** are schematically illustrated, and illustrated as if the entire circumference is connected; however, actually, discontinuous portions exist in a part of the coil pieces **17** in the circumferential direction. The coil pieces **17** have two end portions that are separated across discontinuous portions. Coil pieces **17** that are adjacent in the circumferential direction are connected in series by connecting one end portion of each with a conductor such as printed wiring (not illustrated). These aspects are the same for the second detection coil **6** to the fourth detection coil **8** below.
(41) As illustrated in FIG. **5**B, the second detection coil **6** includes a plurality of coil pieces **18** arranged side by side at equal pitches in the circumferential direction. These coil pieces **18** have a parallelogram shape when viewed from the radial direction, and include wiring that is inclined +45° with respect to the axial direction of the intermediate shaft portion **3**. Coil pieces **18** that are adjacent in the circumferential direction are connected in series by a conductor such as printed wiring (not illustrated).
(42) As illustrated in FIG. **5**D, the third detection coil **7** includes a plurality of coil pieces **19** arranged side by side at equal pitches in the circumferential direction. These coil pieces **19** have a parallelogram shape when viewed from the radial direction, and include wiring that is inclined −45° with respect to the axial direction of the intermediate shaft portion **3**. Coil pieces **19** that are adjacent in the circumferential direction are connected in series by a conductor such as printed wiring (not illustrated).
(43) As illustrated in FIG. **5**C, the fourth detection coil **8** includes a plurality of coil pieces **20** arranged side by side at equal pitches in the circumferential direction. These coil pieces **20** have a parallelogram shape when viewed from the radial direction, and include wiring that is inclined +45° with respect to the axial direction of the intermediate shaft portion **3**. Coil pieces **20** that are adjacent in the circumferential direction are connected in series by a conductor such as printed wiring (not illustrated).
(44) However, when carrying out the present disclosure, configuration regarding the specific shape and arrangement of the first detection coil, the second detection coil, the third detection coil, and the fourth detection coil is not limited to the configuration of this example, and various conventionally known configurations may be adopted.

(45) In this example, as illustrated in FIG. **3**, the first resistance element **9**, the second resistance element **10**, the third resistance element **11**, and the fourth resistance element **12** are attached by soldering, welding, or the like to part of the flexible substrate **16** (the upper part in the left-right direction central part of FIG. **3**). That is, in this example, the flexible substrate **16** of the coil unit **13** has a configuration that makes it possible to attach the first resistance element **9**, the second resistance element **10**, the third resistance element **11**, and the fourth resistance element **12** that are connected to the four sides of the bridge circuit **4**.
(46) In this example, as illustrated in FIG. **6**, a resistor element RE**1** is used for each of the resistance elements **9** to **12**. The resistance element RE**1** has a configuration in which the resistance value between both end points X**1** and X**2**, which is the element's own resistance value, can be adjusted by changing the path length of the copper wire connecting the end points X**1** and X**2** on both sides. That is, the resistance element RE**1**, between the end points X**1** and X**2** on both sides, has a terminal S**1** and a plurality of mating terminals (three terminals S**2**, S**3**, S**4** in the illustrated example) that can be connected to the terminal S**1** by soldering, welding, or the like.
(47) In the resistance element RE**1**, it is possible to change the path length of copper wire connecting the end points X**1** and X**2** on both sides depending on which mating terminal (S**2**, S**3**, S**4**) is connected to terminal S**1**.
(48) However, when carrying out the present disclosure, it is also possible to alternatively use a resistance element RE**2** such as illustrated in FIG. **7** as each of the resistance elements **9** to **12**. The resistance element RE**2** has a plurality of parallel copper wires W connected in parallel in the middle of the copper wiring connecting the end points X**1** and X**2** on both sides. In the resistance element RE**2**, it is possible to change the resistance value between the end points X**1** and X**2** on both sides, which is the element's own resistance, by cutting a part of the parallel copper wires W out of a plurality of parallel copper wires W, for example, a portion surrounded by the dashed line in FIG. **7**.
(49) More specifically, the resistance element RE**2** has a configuration in which it is possible to adjust the element's own resistance value by changing the number of parallel copper wires W to be cut, in other words, by changing the number of uncut parallel copper wires W.
(50) Note that when carrying out the present disclosure, it is also possible to alternatively use resistance elements having a configuration that differs from that of the resistance elements RE**1** (FIG. **6**) and RE**2** (FIG. **2**) and in which it is possible to adjust the resistance value.
(51) In any case, in this example, the resistance values of the resistance elements **9** to **12** forming the bridge circuit **4** are adjusted when the torque measuring device **1** is manufactured, and more specifically, before the resistance elements **9** to **12** are attached to the flexible substrate **16**. In addition, attachment of the resistance elements **9** to **12** to the flexible substrate **16**, as illustrated in FIG. **3**, for example, is performed in a state before the flexible substrate **16** is rolled into a cylindrical shape.
(52) Note that when carrying out the present disclosure, resistance elements whose resistance values cannot be adjusted may also be used instead of the resistance elements **9** to **12**. In this case, a plurality of resistance elements with different resistance values are prepared. As the resistance elements **9** to **12**, resistance elements having required resistance values are selected, and the selected resistance elements are attached to the flexible substrate **16**.
(53) The back yoke **14** is a member that forms a magnetic path for magnetic fluxes generated by the first detection coil **5**, the second detection coil **6**, the third detection coil **7**, and the fourth detection coil **8**. The back yoke **14** is made of a magnetic material such as mild steel and is formed in a cylindrical shape as a whole. The back yoke **14** is arranged coaxially around the coil unit **13**.
(54) In this example, in this state, a clearance in the radial direction is provided along the axial direction between the coil unit **13** and the back yoke **14**. That is, an outer peripheral surface of the coil unit **13** and an inner peripheral surface of the back yoke **14** are separated in the radial direction.
(55) The holder **15** is a member that holds the coil unit **13** and the back yoke **14**, and is made of a

non-magnetic material other than metal, such as synthetic resin, and has an annular shape as a whole. In this example, the holder **15** has a cylindrical holder cylindrical portion **21**, a first outward-facing flange **22** extending outward in the radial direction over the entire circumference from an end portion on one side in the axial direction (left side in FIG. **1**) of the holder cylindrical portion **21**, and a second outward-facing flange **23** extending outward in the radial direction over the entire circumference from an end portion on the other side in the axial direction (right side in FIG. **1**) of the holder cylindrical portion **21**.
(56) The outer diameter of the first outward-facing flange **22** is larger than the outer diameter of the second outward-facing flange **23**. The holder **15** is supported and fixed to the casing while being arranged coaxially around the intermediate shaft portion **3**.
(57) In this example, the coil unit **13** is fitted onto an intermediate portion in the axial direction of the holder cylindrical portion **21**, that is, in a portion of the holder cylindrical portion **21** that is positioned in the axial direction between the first outward-facing flange **22** and the second outward-facing flange **23**.
(58) An end surface of the back yoke **14** on the one side in the axial direction is in contact with an intermediate portion in the radial direction of a side surface on the other side in the axial direction of the first outward-facing flange **22**, and an inner peripheral surface of an end portion on the other side in the axial direction of the back yoke **14** is fitted onto an outer peripheral surface of the second outward-facing flange **23** by an interference fit or the like.
(59) Note that when carrying out the present disclosure, a configuration different from that of this example can be employed for the coil unit that includes the first detection coil **5**, the second detection coil **6**, the third detection coil **7**, and the fourth detection coil **8** that constitute a bridge circuit **4**; the back yoke arranged around the coil unit; the holder that holds the coil unit and back yoke; and the like.
(60) In addition, the first resistance element **9**, the second resistance element **10**, the third resistance element **11**, and the fourth resistance element **12** of the bridge circuit **4** may also be attached to a member different from the coil unit.
(61) When using the torque measuring device **1** of this example, in the bridge circuit **4**, the oscillator applies the input voltage Vi between the two end points, point A and point C, to supply an alternating current to the first detection coil **5**, the second detection coil **6**, the third detection coil **7**, and the fourth detection coil **8**. Then, as indicated by arrows α**1**, α**2**, α**3**, and α**4** in FIG. **5**A to FIG. **5**D, in the first detection coil **5**, the second detection coil **6**, the third detection coil **7**, and the fourth detection coil **8**, currents flow in mutually opposite directions between pairs of coil pieces **17**, **18**, **19**, and **20** adjacent in the circumferential direction.
(62) In other words, pairs of coil pieces **17**, **18**, **19**, and **20** adjacent in the circumferential direction are connected to each other so that the currents flow in such directions. As a result, an alternating magnetic field is generated around the first detection coil **5**, the second detection coil **6**, the third detection coil **7**, and the fourth detection coil **8**, and part of the magnetic flux of this alternating magnetic field passes through a surface layer portion of the intermediate shaft portion **3**.
(63) In this state, when a torque T in a direction indicated by arrow CW in FIG. **1** is applied to the intermediate shaft portion **3**, a tensile stress (+σ) in a +45° direction with respect to the axial direction and a compressive stress (−σ) in a −45° direction with respect to the axial direction act on the rotating shaft **2**. Then, due to an inverse magnetostriction effect, the magnetic permeability of the intermediate shaft portion **3** increases in the +45° direction, which is the direction in which the tensile stress (+σ) acts, and the magnetic permeability of the intermediate shaft portion **3** decreases in the −45° direction, which is the direction in which the compressive stress (−σ) acts.
(64) On the other hand, the first detection coil **5** and the third detection coil **7** are configured to include wiring inclined −45° with respect to the axial direction of the intermediate shaft portion **3**, and part of the magnetic flux of the alternating magnetic field generated around the wiring passes through the surface layer of the intermediate shaft portion **3** in the +45° direction, which is the

direction in which the magnetic permeability increases. Therefore, the inductances L**1** and L**3** of the first detection coil **5** and the third detection coil **7** increase, respectively.
(65) Moreover, the second detection coil **6** and the fourth detection coil **7** are configured to include wiring inclined +45° with respect to the axial direction of the intermediate shaft portion **3**, and part of the magnetic flux of the alternating magnetic field generated around the wiring passes through the surface layer of the intermediate shaft portion **3** in the −45° direction, which is the direction in which the magnetic permeability decreases. Therefore, the inductances L**2** and L**4** of the second detection coil **6** and the fourth detection coil **8** decrease, respectively.
(66) In contrast, when a torque T in the direction indicated by the arrow CCW in FIG. **1** is applied to the intermediate shaft portion **3**, by the action opposite to the case described above, the inductances L**1** and L**3** of the first detection coil **5** and the third detection coil **7** decrease, and the inductances L**2** and L**4** of the second detection coil **6** and the fourth detection coil **8** increase.
(67) In any case, in the bridge circuit **4**, an output voltage Vo corresponding to the direction and magnitude of the torque T applied to the rotating shaft **2** is obtained as the voltage between the two midpoints, point B and point D. Therefore, when this output voltage Vo is measured by the voltage measuring portion, the direction and magnitude of the torque T can be obtained based on the measured output voltage Vo.
(68) With the torque measuring device **1** of this example, it is possible to suppress changes in the output voltage Vo of the bridge circuit **4** due to temperature changes.
(69) That is, the resistance values R**1**, R**2**, R**3**, and R**4** of the four sides of the bridge circuit **4** increase and decrease with temperature change, and the output voltage Vo also increases and decreases with the increase and decrease. At this time, the amount of change in the output voltage Vo with respect to the temperature change increases as the variation in the resistance values R**1**, R**2**, R**3**, and R**4** on the four sides increases. Here, when taking into consideration the ratio (R**1**×R**3**)/(R**2**×R**4**), the closer the ratio is to 1, the amount of change in the output voltage Vo becomes small with respect to the temperature change.
(70) In this regard, in this example, the resistance values of resistance elements **9** to **12** connected to each of the four sides are adjusted so that the ratio (R**1**×R**3**)/(R**2**×R**4**) approaches 1 compared to the case where the resistance elements **9** to **12** are not connected to the four sides. Therefore, it is possible to suppress the change in the output voltage Vo of the bridge circuit **4** due to the temperature change.
Second Example
(71) A second example of an embodiment of the present disclosure will be described with reference to FIG. **8**.
(72) In this example, a bridge circuit **4***a* has a resistance element **24** connected to only one of four sides. Note that in FIG. **8**, the resistance element **24** is connected to a side where a first detection coil **5** exists; however, this is merely an example. In this example, in the bridge circuit **4***a*, a resistance value of the resistance element **24** is adjusted so that a ratio (R**1**×R**3**)/(R**2**×R**4**) approaches 1 compared to when the resistance element **24** is not connected.
(73) For this reason, in this example, before connecting the resistance element **24** to one of the four sides of the bridge circuit **4***a*, resistance values of respective resistance components r**1**, r**2**, r**3**, and r**4** of the first detection coil **5**, the second detection coil **6**, the third detection coil **7**, and the fourth detection coil **8** are measured with a tester. The resistance values of the resistance components r**1**, r**2**, r**3**, and r**4** that are measured in this way, and the resistance value of the resistance element **24** are combined, and the resistance value of the resistance element **24** is adjusted so that the ratio (R**1**×R**3**)/(R**2**×R**4**) approaches 1.
(74) In this example, of the detection coils **5** to **8** on the four sides, the resistance element **24** is connected to the side where the detection coil having the smallest resistance value of the resistance components r**1**, r**2**, r**3**, and r**4** measured by the tester exists. For example, in a case where the detection coil with the smallest resistance value of the resistance components measured by the

tester is the first detection coil **5**, as illustrated in FIG. **8**, the resistance element **24** is connected to the side where the first detection coil **5** exists. At this time, the resistance value of the resistance element **24** is adjusted so that the ratio (R**1**×R**3**)/(R**2**×R**4**) approaches 1 before connecting the resistance element **24** to the side of the bridge circuit **4***a*.
(75) Other configurations and effects are the same as those of the first example.
(76) When carrying out the present disclosure, it is also possible to adopt a configuration in which resistance elements for adjusting the resistance value of the sides are connected only to any two or three sides of the four sides of the bridge circuit. In these cases as well, by adjusting the resistance values of the respective resistance elements on the two or three sides, the ratio (R**1**×R**3**)/(R**2**×R**4**) can be made to approach 1 compared to when these resistance elements are not connected.
(77) Whereas many alterations and modifications of the present disclosure will no doubt become apparent to a person of ordinary skill in the art after having read the foregoing description, it is to be understood that the particular embodiments shown and described by way of illustration are in no way intended to be considered limiting.
(78) It is noted that the foregoing examples have been provided merely for the purpose of explanation and are in no way to be construed as limiting of the present disclosure. While the present disclosure has been described with reference to exemplary embodiments, it is understood that the words, which have been used herein, are words of description and illustration, rather than words of limitation. Changes may be made, within the purview of the appended claims, as presently stated and as amended, without departing from the scope and spirit of the present disclosure in its aspects. Although the present disclosure has been described herein with reference to particular means, materials and embodiments, the present disclosure is not intended to be limited to the particulars disclosed herein; rather, the present disclosure extends to all functionally equivalent structures, methods and uses, such as are within the scope of the appended claims.
REFERENCE SIGNS LIST
(79) **1** Torque measuring device **2** Rotating shaft **3** Intermediate shaft portion **4**, **4***a* Bridge circuit **5** First detection coil **6** Second detection coil **7** Third detection coil **8** Fourth detection coil **9** First resistance element **10** Second resistance element **11** Third resistance element **12** Fourth resistance element **13** Coil unit **14** Back yoke **15** Holder **16** Flexible substrate **17** Coil piece **18** Coil piece **19** Coil piece **20** Coil piece **21** Holder cylindrical portion **22** First outward-facing flange **23** Second outward-facing flange **24** Resistance element **100** Bridge circuit **101** Rotating shaft **102** Magnetostrictive effect section **103** First detection coil **104** Second detection coil **105** Third detection coil **106** Fourth detection coil

Claims

1. A torque measuring device comprising a bridge circuit in which four detection coils arranged around a magnetostrictive effect section of a rotating shaft are arranged on four sides; wherein the bridge circuit includes a resistance element connected to at least one of the four sides, and the resistance element is connected to the at least one of the four sides in a state in which a resistance value of the resistance element is adjusted so that a ratio (R**1**×R**3**)/(R**2**×R**4**) of a product R**1**×R**3** of resistance values R**1** and R**3** of two opposite sides forming one pair of opposite sides of the four sides, and a product R**2**×R**4** of resistance values R**2** and R**4** of two opposite sides forming another pair of opposite sides of the four sides approaches 1, the resistance values R**1**, R**2**, R**3**, and R**4** of the four sides measured in advance before the resistance element is connected to the at least one of the four sides.
2. The torque measuring device according to claim 1, wherein the bridge circuit includes four resistance elements respectively configured by the resistance element and connected to one each of the four sides.
3. The torque measuring device according to claim 1, wherein the resistance element has a

configuration in which the resistance value between end points on both sides of the resistance element, which is the resistance value of the resistance element, is adjustable by changing a path length of copper wire connecting the end points on both sides of the resistance element.
4. The torque measuring device according to claim 1, wherein the resistance element has a configuration in which a plurality of parallel copper wires are connected in parallel in a middle of the copper wiring connecting the end points on both sides of the resistance element, and the resistance value between the end points on both sides, which is the resistance value of the resistance element, is adjustable by changing the number of the plurality of parallel copper wires by cutting a part of the plurality of parallel copper wires.
5. The torque measuring device according to claim 1, further comprising: a coil unit formed in a cylindrical shape and including the four detection coils; a back yoke formed in a cylindrical shape and arranged coaxially around the coil unit; and a holder configured to hold the coil unit and the back yoke.